



<Code Club JUSL/>



DEPARTMENT OF INFORMATION TECHNOLOGY
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D.E. Shaw & Co.

Internship Acceptance Guide

A complete guide for aspiring interns

Student's Experience & Suggestions

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Daniyal Anis

Information Technology

Name	Daniyal Anis
Department	Information Technology
Year of Grad.	2027

Experience

Online Assessment (OA) : It consisted of three sections:

Data Structures & Algorithms (25 minutes) :

One coding problem was provided.

Problem Overview:

Given a list of integers representing a stack of elements (from left to right), the goal was to organize them into groups based on the following rules: Start with the left-most element and place it into a new group.

From the remaining elements on the right, select the first element strictly less than the last added element in the group and add it. Repeat until no more elements can be added to the current group. Remove all grouped elements from the original list and repeat until all elements are grouped.

Return the list of all groups formed.

Difficulty: Easy

CS Fundamentals (20 minutes) :

10 multiple-choice questions covering Operating Systems, Computer Networks, DBMS and Object-Oriented Programming.

Difficulty: Easy.

Aptitude & Quantitative Ability (20 minutes) :

10 questions, with calculations and reasoning-based problems.

Difficulty: Moderate to Difficult (lengthy in nature).

Shortlist Outcome: Out of all participants, 15 candidates were shortlisted for interviews.

Technical Round 1 (~50 minutes) :

Began with introductions and a brief discussion about the Online Assessment. The interviewer opened up my github and projects, which were listed on the resume. Subsequent questions were asked from each of the projects as well as the projects on Github. Extensive Questions about various coding profiles. Focused heavily on Computer Networks, DBMS, OOPS, and Operating Systems, with conceptual deep-dives. In-depth discussion on resume-listed projects, including:

Tech stack used and reasons for its selection.

Project architecture and implementation details.

Core Computer Science topics covered:

Virtual Methods – Implementation and use cases.

Basic Linux commands for file listing.

Networking concepts such as how ping works, BGP, and ARP.

Multi Threading.

Difference between compiler and interpreter.

Each question was followed by deeper, conceptual cross-questions. Interview style was structured but friendly, focusing on thorough understanding of fundamentals and project depth.

Shortlist Outcome: Out of 15 candidates shortlisted for Round 1, 5 candidates were shortlisted for Technical Round 2 and HR Round.

Technical Round 2 (~1 hour 20 minutes) :

Similar in structure to Round 1 but with a new panel. Focused heavily on Computer Networks and Operating Systems, with conceptual deep-dives. One DSA Problem: Multi-source BFS based problem. You had to explain the time and space complexity of the code. You were provided with A4 sheets and a pen, to write up the solution and explain your approach. I was asked why BFS was used instead of DFS for the problem. They also asked what the time complexity would be if Dijkstra's algorithm had been used instead, and whether that would have been a better approach than the one I had provided. The interviewers were satisfied with my response.

The DSA problem was as follows:

In a city represented as a 2D grid, there are several dark stores, houses, and obstacles. Your task is to determine the minimum time it takes for goods from any dark store to reach each house. Goods can only travel through empty cells, and cannot pass through obstacles or houses. However, they can be delivered to houses by reaching an adjacent cell (up, down, left, or right). Your job is to compute the minimum time required for goods from any dark store to reach each house in the grid.

Example theoretical questions:

What is a proxy and why is it used?

TCP vs UDP – Advantages of one over the other.

How does DNS work?

Questions were designed to test the problem-solving approach and technical thought process. A puzzle was asked to be solved, which was as follows:

You are given 10 coins consisting of 5 of them are heads up and the remaining are tails up. You are blindfolded and you aren't able to make out which coin is head/tails. You have to assemble two stacks of coins, each consisting of five coins, such that the number of heads in both stacks are the same. You are allowed to flip coins any number of times to achieve the goal.

I had 30 seconds to solve the puzzle. I quickly came up with a solution and explained my approach to the interviewers, who were satisfied with it. This marked the end of the interview.

HR Round (~40 minutes) :

Standard HR-format discussion covering:

Introduction and resume review.

Strengths and key achievements.

Location preference.

Understanding of the role after technical interviews.

Interaction was conversational and straightforward. You may ask questions at the end of each round.

Prep. Tips

Strong preparation in Core CS Fundamentals and project-specific knowledge is essential.

They are looking for a well rounded person with extensive experience in competitive programming/DSA as well strong in Core CS Fundamentals. Thorough clarity on reasoning behind technical choices is valued.

Coding questions tested both algorithm design and implementation speed.
The process was well-structured, and interviewers maintained a professional, approachable demeanor throughout.
Be confident about your answer and try to be straight forward with your answers.
If you use a term in your answer, be sure to be well-versed in related matters.

✓ **Selected**

Name	Sayan Gupta
Department	Computer Science & Engineering
Year of Grad.	2027

Experience

OA (08/08/2025) :

It contained 20 MCQs from CS fundamentals (C programming, DBMS, Computer Networks) and one DSA question. The DSA question was: You are given a list of files with some numerical priorities. You must go from left to right and in one stack you can only stack files in decreasing order of priority. How many total stacks will have to be made to accommodate all the files? (Constraints: $1 \leq n \leq 3 \times 10^5$)

Technical Round 1 (08/08/2025) :

It started off with, "Tell me about yourself." I mentioned my interests and projects, among other things. Since my main interest was ML, I was asked questions like, "What is recall?" "Why is accuracy a bad measure?" "What is cosine similarity?" and "What are the different ways of evaluating an ML model?"

I had an Automatic Differentiation Python Library as a project, to which they were very interested, and asked questions like how I came up with the idea of creating such a project, how it works overall (computational graphs), etc.

I had a project using LLMs, so they asked me about the architecture of an LLM, what word embeddings are, and how similarities between words are measured. Another project of mine used FastAPI, so they asked why I used FastAPI and then moved on to questions about async/await, what threading is, how to do threading in Python, and how it's different from async/await. I was able to answer all of them correctly.

Then they moved on to Computer Networks, asking about proxies, whether IP addresses can be the same, what happens when you type a link into the browser, etc.

At last, they asked a DSA question: Given a list of nodes of a binary tree and an online query function isAncestor(a, b) which returns whether a is an ancestor of b, I had to reconstruct the entire tree. I first came up with a brute-force solution, and they asked for its time complexity. I wasn't asked to optimize it further or code the solution.

Finally, they asked if I had any questions. I asked about what projects are given during the internship, what languages are used or preferred, etc.

HR Round (08/08/2025):

I was immediately called for the HR round and told that I was selected for technical round 2. This round began by asking about the online assessment, specifically whether I found it easy or hard. I told her it was easy, so she asked why I thought that, and asked about my coding profiles (I have Codeforces and CodeChef). Other questions followed, like: "Why did you apply here?", "Do you think you will be a good fit for the role and why?", "What are your biggest achievements?", "How do you feel college is different from school?", "What have you learned in your two years of college?", "What do you do outside of academics?", "What challenges have you faced when working in a group?", etc. The round was more like a discussion than an interview.

Technical Round 2 (09/08/2025):

This round immediately started with a DSA question: You are given an array of numbers and a value k, and for all windows of length k in order from beginning to end, you have to return the element which appears the maximum number of times for each window. I was also asked to code the solution (in any language of choice, I chose Python).

Then they asked what rate limiters are, why they are used and how I would implement one from scratch. I mentioned using Redis with a defined timeout, so they started talking about Redis, what it is and why it's used. I was then asked to design it without using the timeout feature.

They then moved to fundamental Python questions (my main expertise is Python), like: "What is a decorator and why is it used?", "Create a decorator to measure the time taken by a function to run.", "What is perf_counter?", "What are *args and **kwargs?", "What is GIL and why is it even applied?". I was asked about event loops and system calls.

The very last question was, "What are database indexes and why are they used?". I mentioned B-trees, so they asked me to draw and explain the structure of a B-tree, which I was able to show correctly, and the interview ended.

Prep. Tips

- 1) Focus on topics like Computer Networks, Operating Systems, Concurrency besides just DSA, as these are asked rigourously.
- 2) Come prepared to implement things from scratch, so read the internal workings of everything on your resume well.

Addit. Notes

- 1) Know your Resume (skills/projects) well, do not put anything on the resume which you are not sure of.
- 2) If you do not know something just tell the interviewer you don't know. Do not try to bluff anything.
- 3) Always talk out your entire approach loudly for DSA questions, designing questions, etc.

✓ Selected
